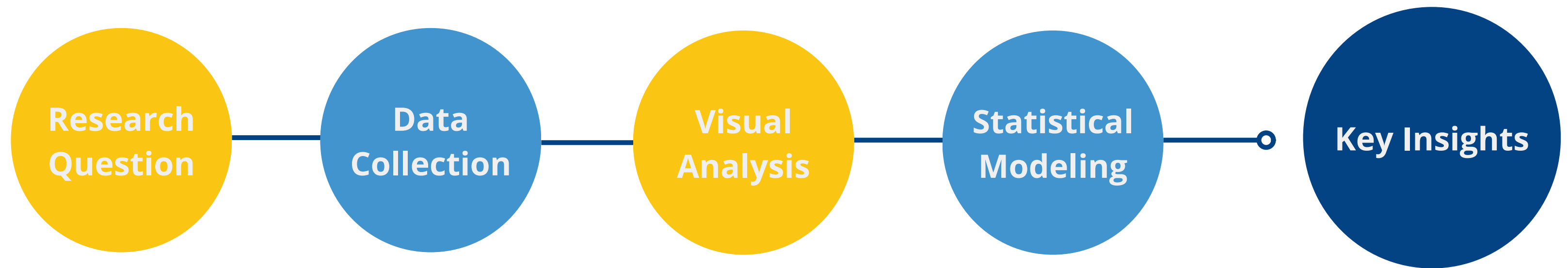


ANALYZING TOP BRAWL STARS PLAYERS: *THE BEST AMONG THE BEST*

Presented by Zhamanta Chavez



AGENDA





QUESTION

What patterns or playstyles distinguish top-tier Brawl Stars players from others within the top 200?

Why it is important:

- Inform decisions for esports design
- Guide game balance adjustments
- Suggest new playtesting objectives



DATA VARIABLES

TROPHIES

The number of trophies a player has; earned or lost when winning or losing games. Determines player rank.

3V3 WINS

Total number of wins in 3v3 game modes.

SOLO WINS

Total number of wins in solo game modes.

DUO WINS

Total number of wins in Duo Showdown.

BEST BRAWLER

The player's highest-trophy brawler, representing their most-used or strongest character.

BEST BRAWLER TYPE

The class/role of the player's best brawler (e.g., Damage Dealer, Tank, Support).

200
avors

Brawl Stars API.

EP/14/01/2020	161	111111	28595	4000	2512	SHELLY	Damage Dealer
EP/14/01/2020	162	111111	14606	5636	4071	NIITA	Damage Dealer
EP/14/01/2020	163	111111	12065	6618	1208	NIIMA	Unknown
EP/14/01/2020	164	111111	29370	4168	19062	BROCK	Marksmen
EP/14/01/2020	165	111111	21894	3251	2385	JIJU	Artillery
EP/14/01/2020	166	111098	71746	4594	5393	BIBI	Tank
EP/14/01/2020	167	111097	32540	7484	1525	SHELLY	Damage Dealer
EP/14/01/2020	168	110669	21690	6548	2391	IAE-YONG	Unknown
EP/14/01/2020	169	111031	4710	3686	1728	SHELLY	Damage Dealer
EP/14/01/2020	170	110978	22562	1572	1330	CORDELIUS	Assassin
EP/14/01/2020	171	110951	14180	5005	5105	HANK	Tank
EP/14/01/2020	172	110939	17256	1017	1470	COLT	Damage Dealer
EP/14/01/2020	173	110933	18599	1813	7662	IAE-YONG	Unknown
EP/14/01/2020	174	110909	9821	11571	6049	CORDELIUS	Assassin
EP/14/01/2020	175	110893	16851	15752	3679	EMZ	Controller
EP/14/01/2020	176	110834	36736	6709	5280	PAM	Support
EP/14/01/2020	177	110848	55915	3845	1040	SHELLY	Damage Dealer
EP/14/01/2020	178	110831	39919	2382	7768	HANK	Tank
EP/14/01/2020	179	110844	75949	3589	20616	BO	Controller
EP/14/01/2020	180	110776	10649	4559	7882	HANK	Tank

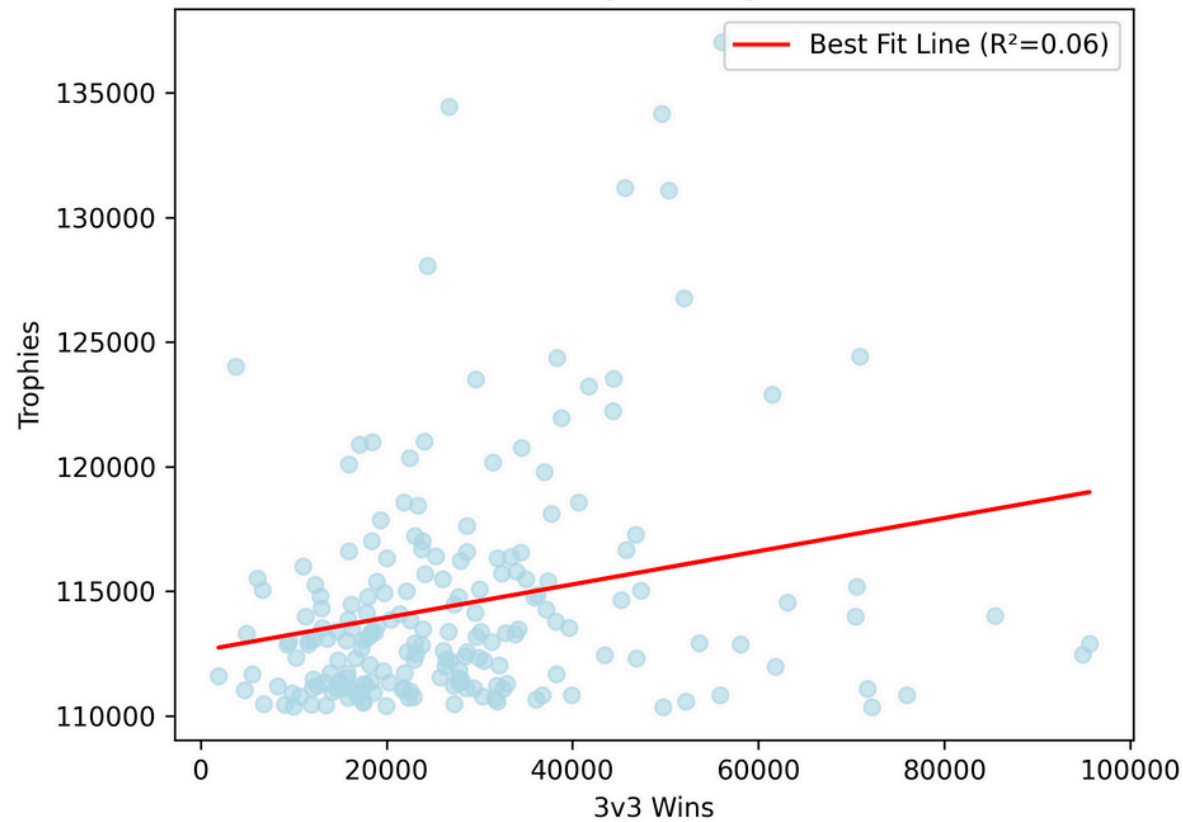
[illegible]

A	B
Name	Tag
SnakeThug	#CGQCYJGY
KiraVGX	#P2U2GRQL
Kenda	#RQQ08UV
Almighty GodðŸ%o	#29CVP0V9
Uncle X	#8G2BPPG

Statistics for the first 20

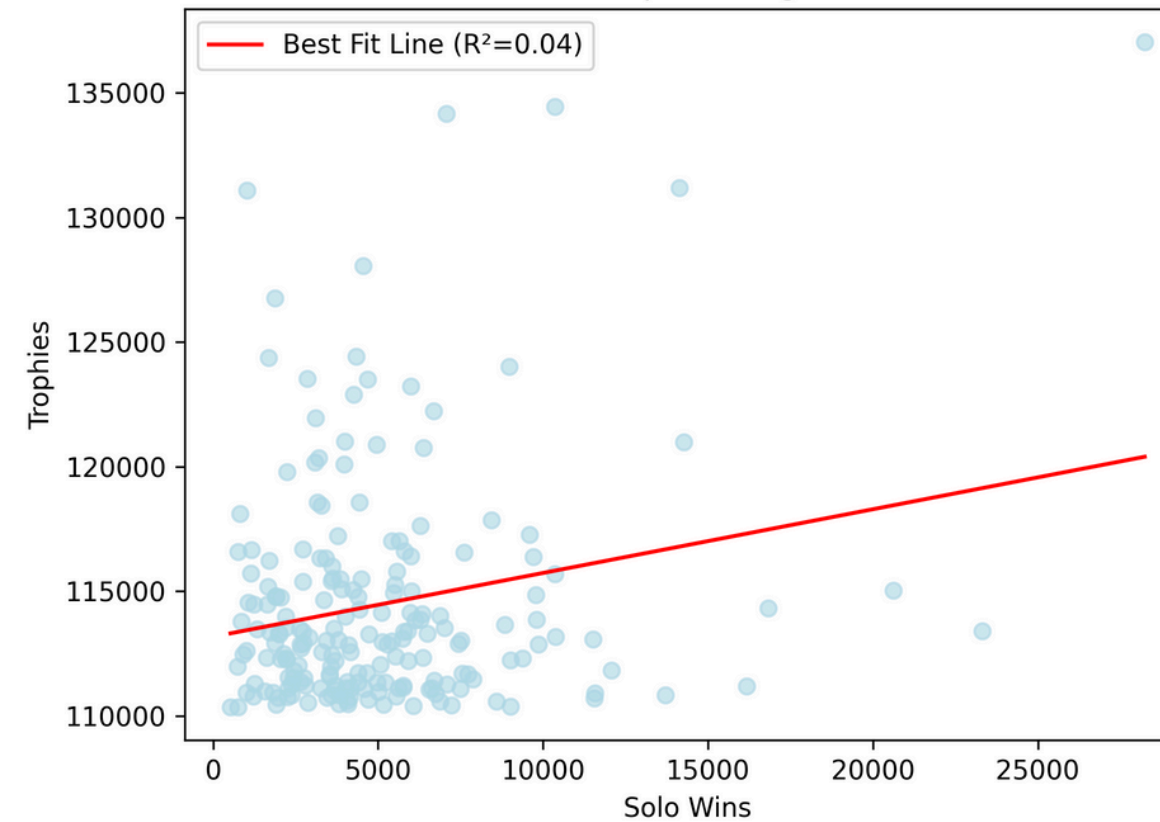
MATCH TYPE CORRELATIONS

3v3 Wins vs Trophies Regression Line



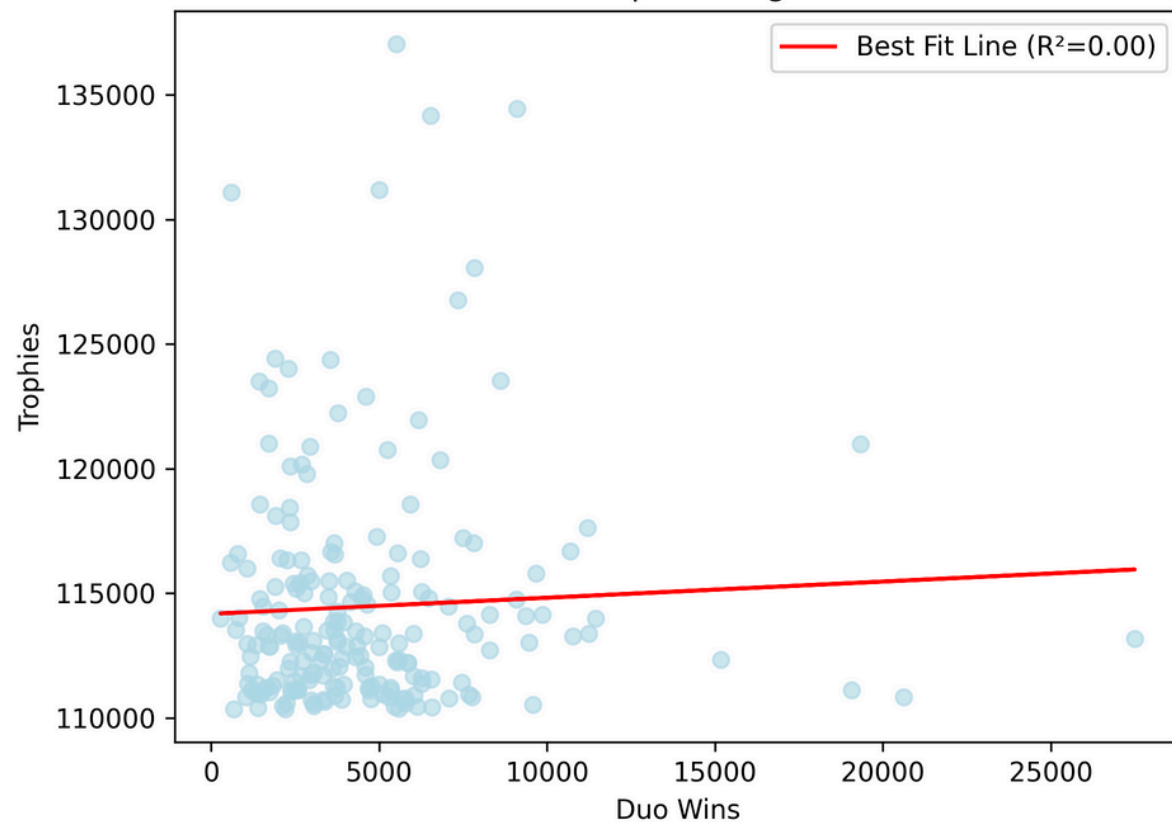
$R^2 = 0.06$

Solo Wins vs Trophies Regression Line



$R^2 = 0.04$

Duo Wins vs Trophies Regression Line



$R^2 = 0.00$

MISLEADING CORRELATION

- Concluding that “more 3v3 wins result in more trophies” is dangerous as these results may be mainly attributed to more time played
- A player with more total matches played (more experience) could naturally have more trophies

DUO WINS LACK IMPACT

What can be safely concluded is that the **count of Duo Wins does not have an impact** on how many trophies you have regardless of experience.

REGRESSION MODEL AND PERMUTATION IMPORTANCE

OLS Regression Results

=====						
Dep. Variable:	Trophies		R-squared:	0.110		
Model:	OLS		Adj. R-squared:	0.096		
Method:	Least Squares		F-statistic:	8.059		
Date:	Tue, 25 Nov 2025		Prob (F-statistic):	4.33e-05		
Time:	01:32:50		Log-Likelihood:	-1960.9		
No. Observations:	200		AIC:	3930.		
Df Residuals:	196		BIC:	3943.		
Df Model:	3					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	1.11e+05	825.861	134.443	0.000	1.09e+05	1.13e+05
3v3Victories	0.0716	0.019	3.785	0.000	0.034	0.109
SoloVictories	0.2792	0.083	3.363	0.001	0.115	0.443
DuoVictories	0.0069	0.089	0.077	0.939	-0.170	0.183
=====						
Omnibus:	76.175		Durbin-Watson:	0.218		
Prob(Omnibus):	0.000		Jarque-Bera (JB):	202.686		
Skew:	1.678		Prob(JB):	9.71e-45		
Kurtosis:	6.615		Cond. No.	8.70e+04		
=====						

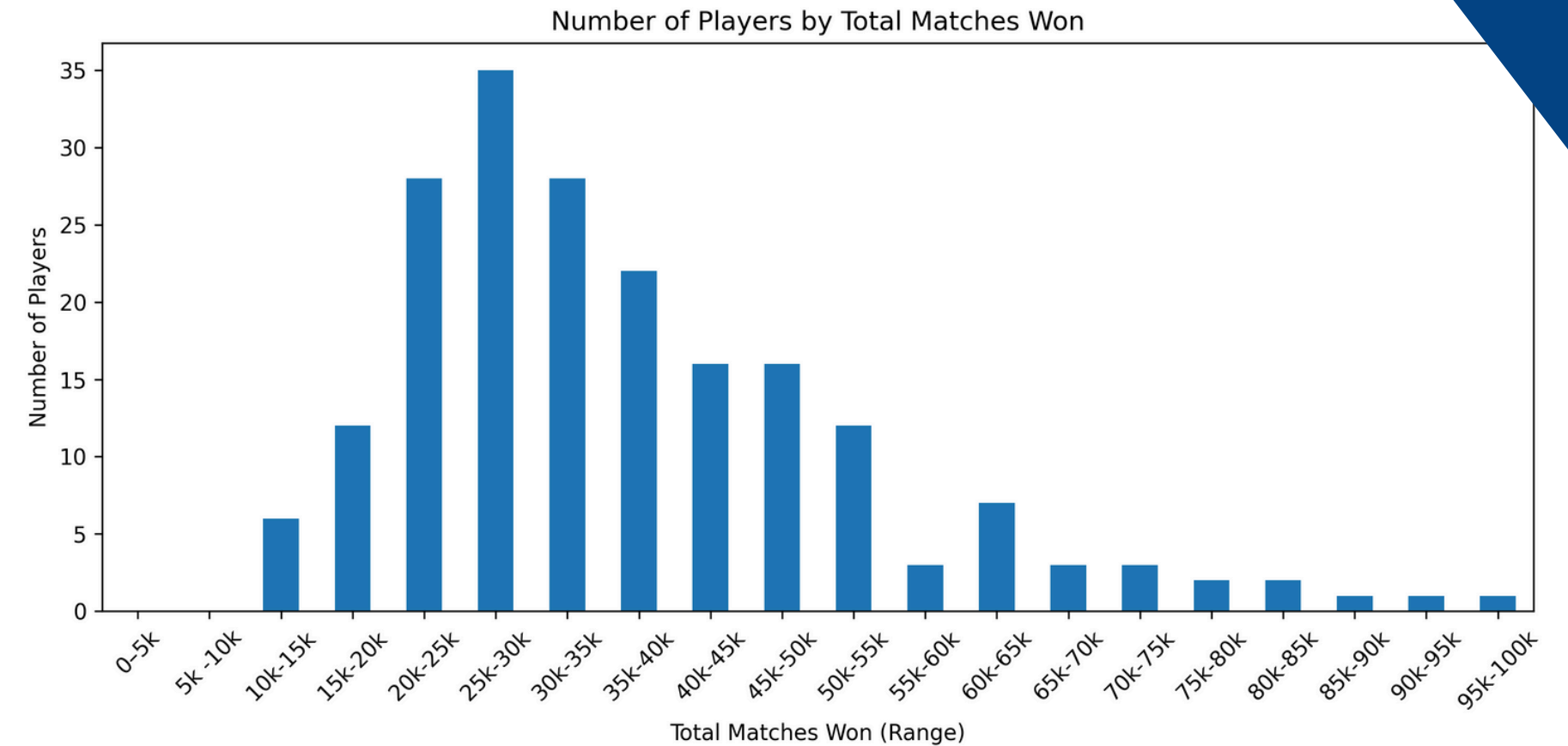
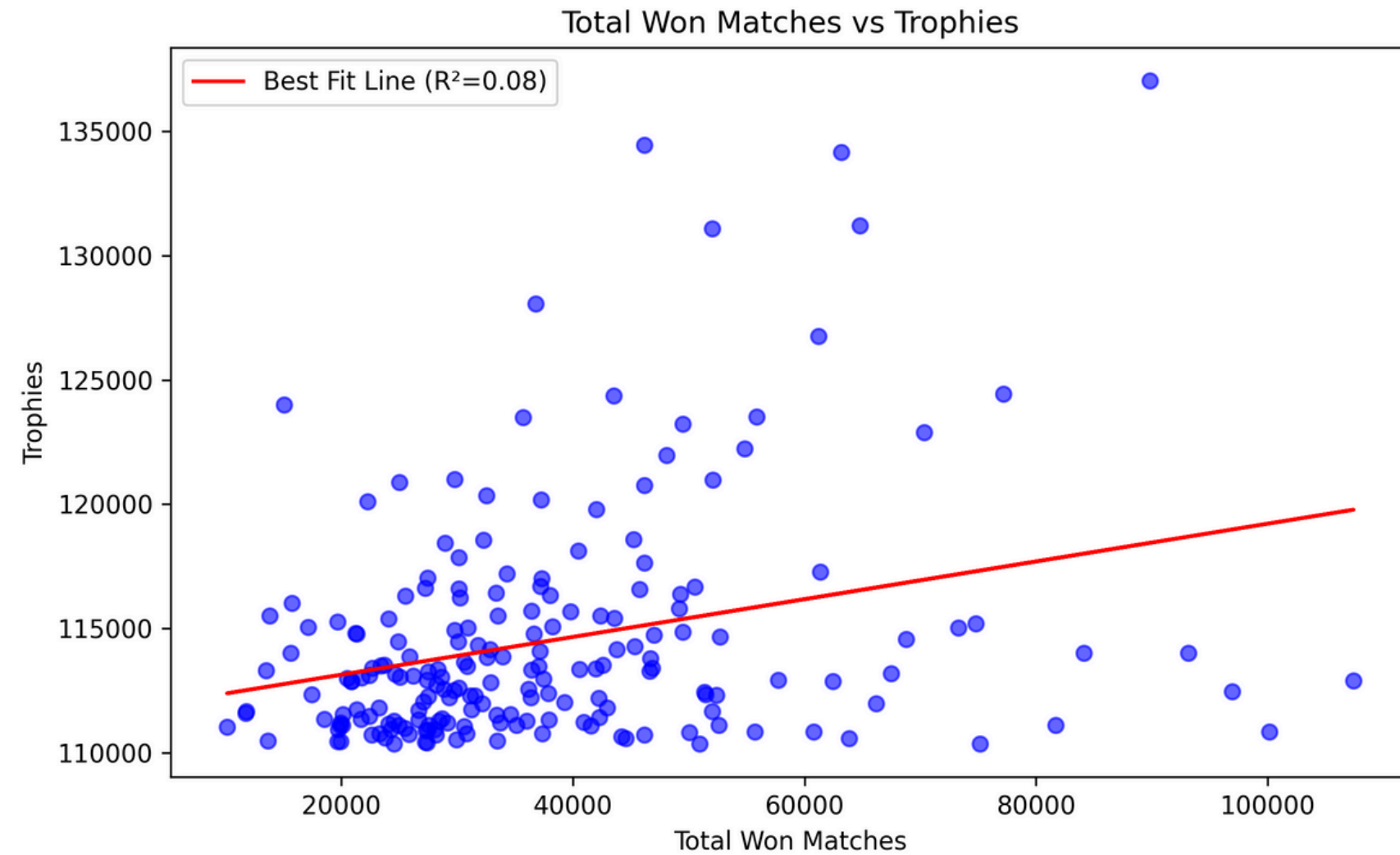
Permutation Importance

feature	importance_mean	importance_std
SoloVictories	0.143419	0.088843
3v3Victories	0.006664	0.105634
DuoVictories	-0.002810	0.004702

KEY FINDINGS

- 11% of the variability in trophies can be explained by 3v3, Solo, and Duo Victories
- Only the 3v3 and Solo Victories are statistically significant
- Solo Victories have the biggest impact on trophy count among the match types
- Shuffling Solo Victories hurts the model's accuracy the most
- 89% of the variability in trophies is likely explained by external factors, such as playtime

STRATIFICATION



TOTAL WON MATCHES

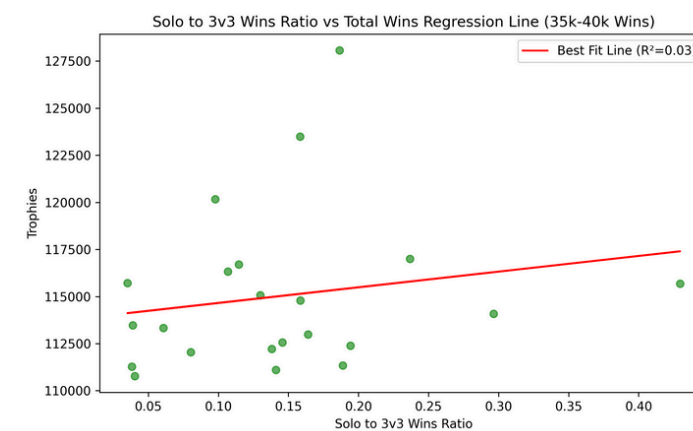
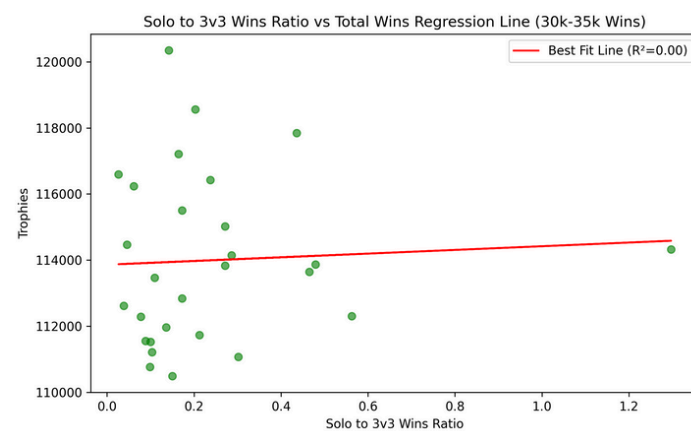
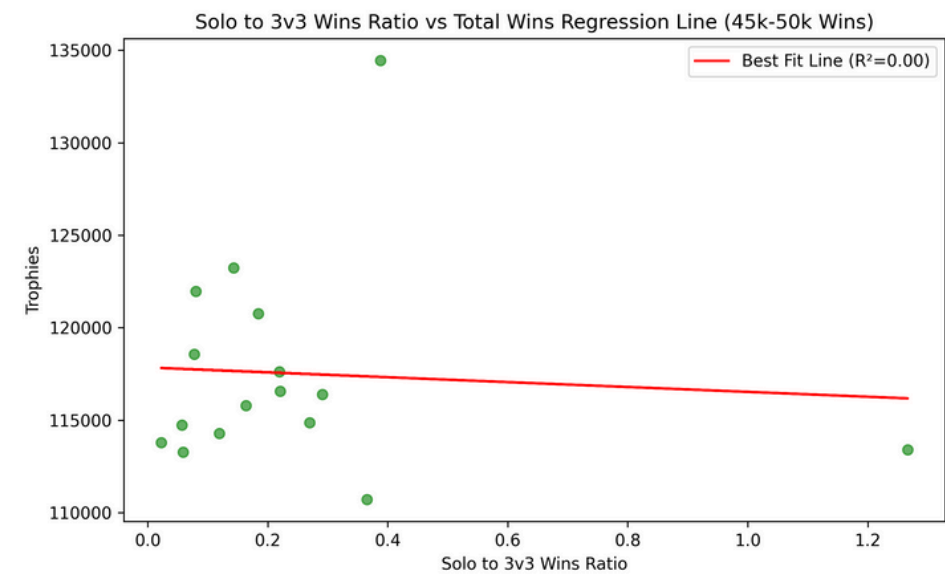
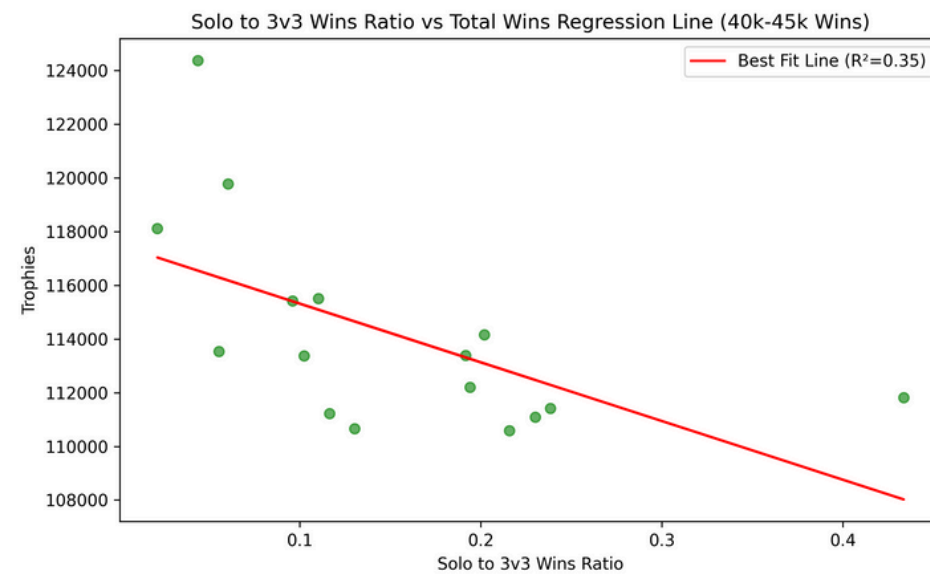
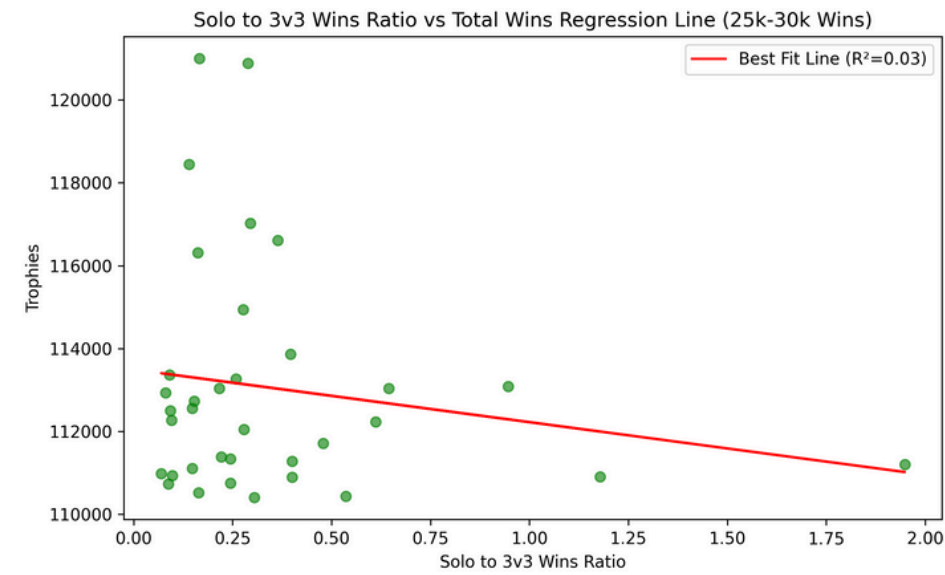
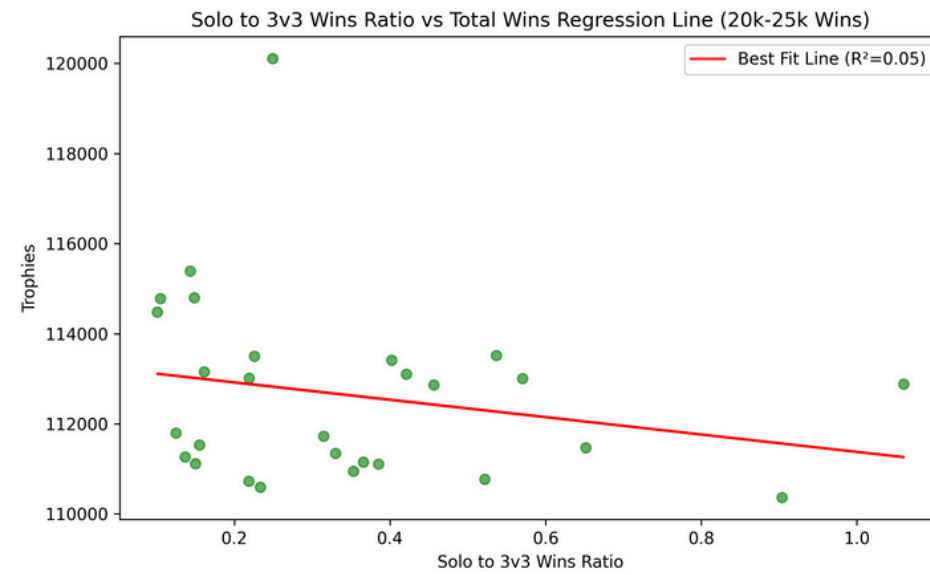
- The number of total won matches may be the closest indicator to a player's experience that is present in the data collected
- Total matches (won + lost matches) would be ideal

Players with high total won matches but low trophy count may have a poor win rate.

STRATIFICATION

- Grouping players by 5k wins may slightly address the experience gaps
- It is safer to analyze groups between 20k to 50k wins as the samples are sufficiently large

SOLO WINS TO 3V3 WINS RATIOS



STRATIFIED RESULTS

Players that win many more 3v3 matches than solo matches tend to have higher trophies, best represented by players who have a win count between 40k and 45k, whose correlation resulted in an R^2 of .35.

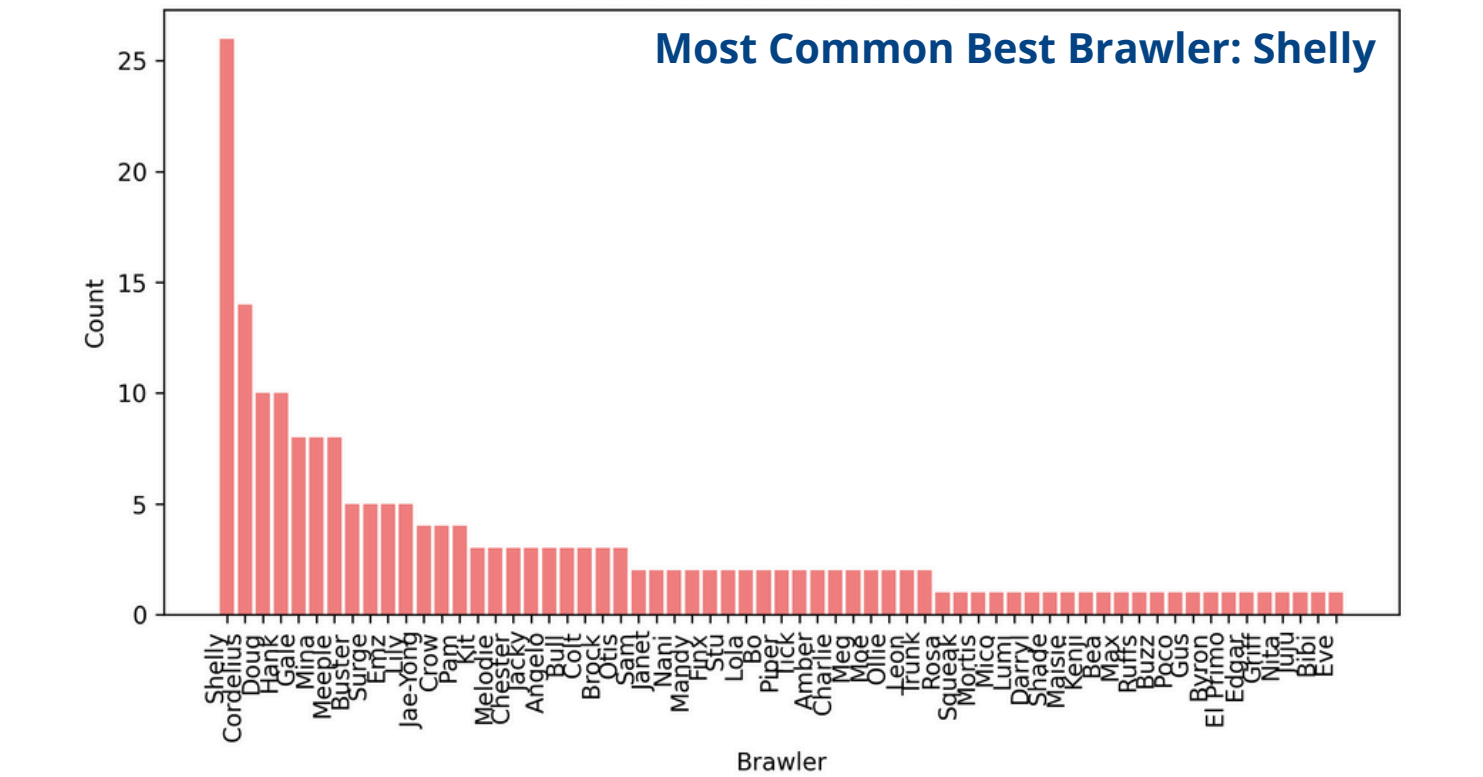
POSITIVE CORRELATION GROUPS

- Despite an R^2 of 0, players with a win count between 30k and 35k may be showing a slightly positive correlation due to an outlier with a ratio higher than 1.2
- Players with a win count between 35k and 40k, despite their positive correlation, all have ratios below .5

BRAWLER AND BRAWLER TYPE BAR GRAPHS

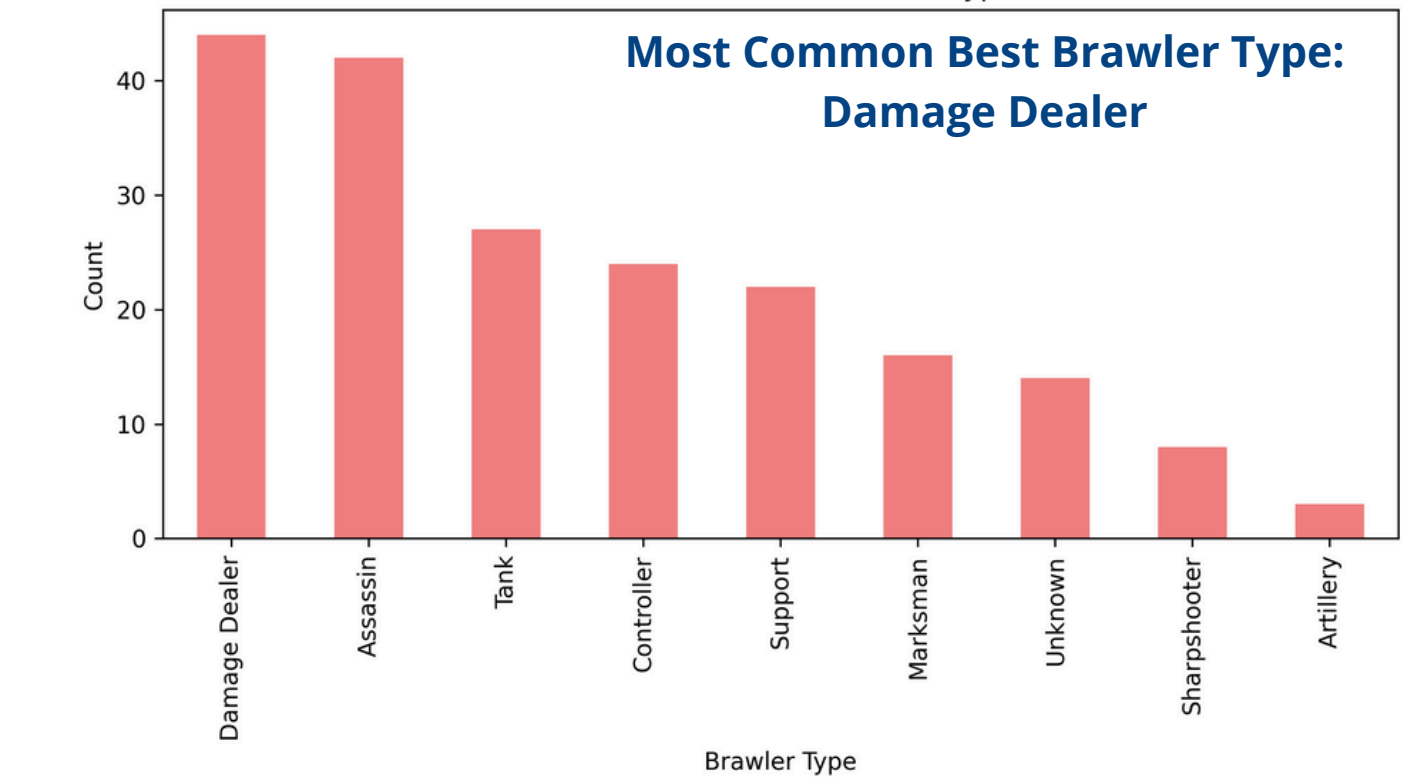
Most Common Best Brawler

Most Common Best Brawler: Shelly



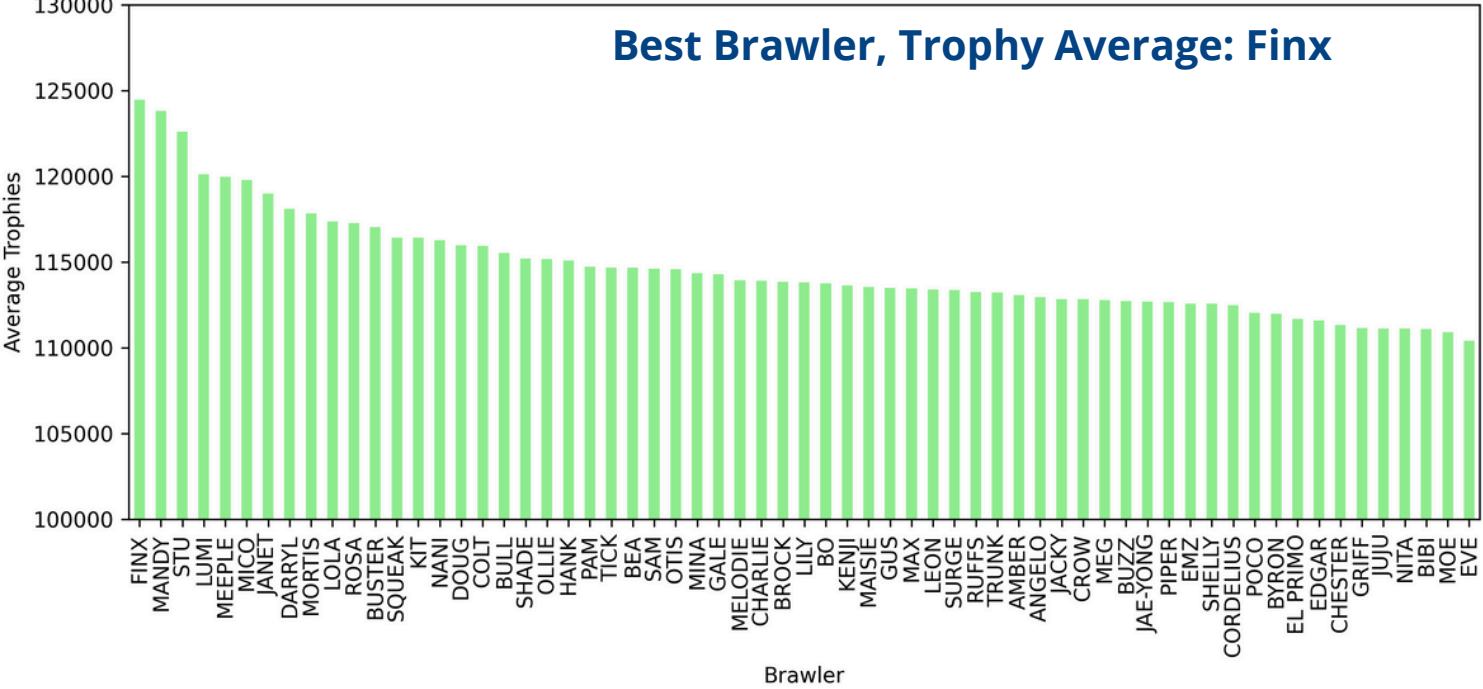
Most Common Best Brawler Type

Most Common Best Brawler Type:
Damage Dealer



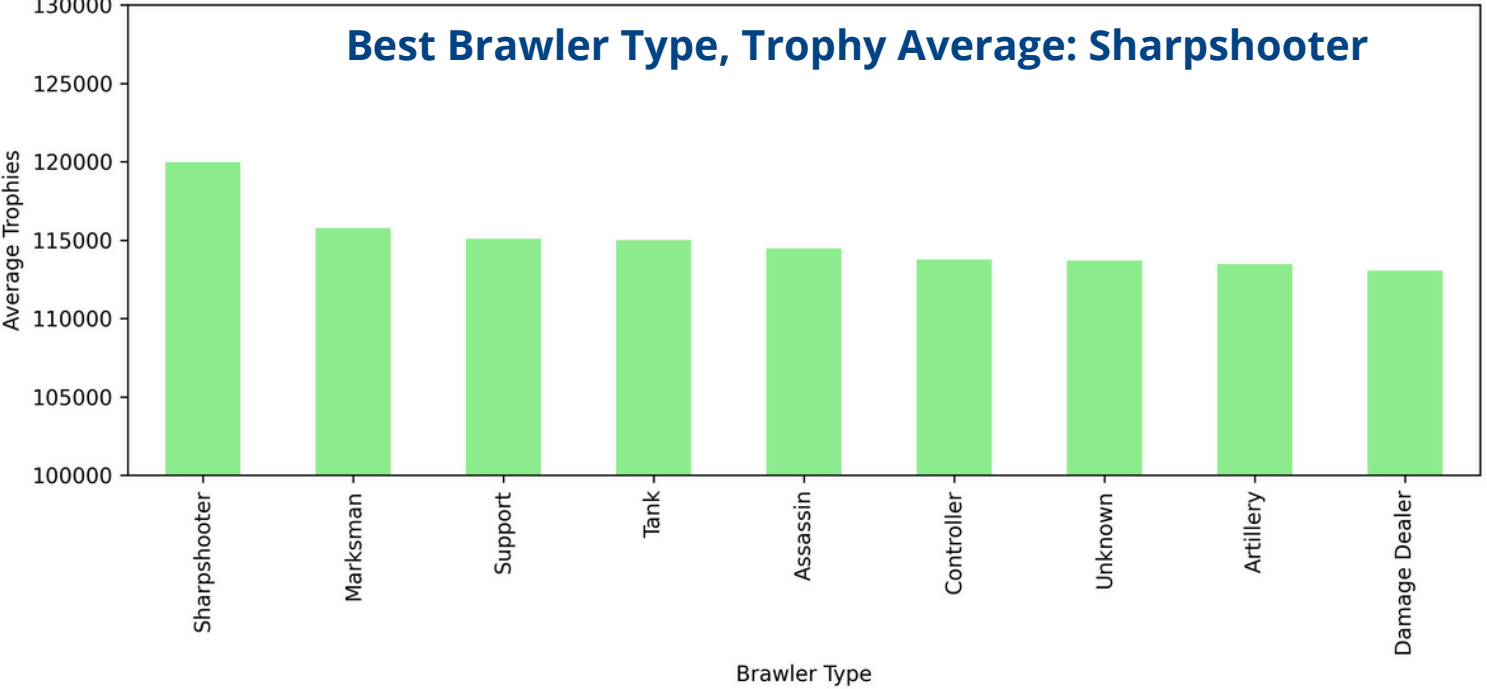
Average Trophies by Best Brawler

Best Brawler, Trophy Average: Finx



Average Trophies by Best Brawler Type

Best Brawler Type, Trophy Average: Sharpshooter



AVERAGING

Averaging trophies per brawler or brawler type may be punishing brawlers or brawler types that are evenly distributed across the table, as is the case with Damage Dealers.

ANOVA MODEL AND KRUSKAL-WALLIS TEST

1. Anova Model

	sum_sq	df	F	PR(>F)	eta_sq
C(BestBrawler)	1.382177e+09	62.0	1.041289	0.415229	0.320301
Residual	2.933061e+09	137.0	NaN	NaN	0.679699

LOW STATISTICAL POWER

The variable Best Brawler being statistically insignificant may be due to uneven sample sizes of players across brawlers, rather than the absence of their effect on trophy count.

	sum_sq	df	F	PR(>F)	eta_sq
C(BestBrawlerType)	3.977113e+08	8.0	2.423814	0.016228	0.092164
Residual	3.917526e+09	191.0	NaN	NaN	0.907836

VIOLATION OF ANOVA ASSUMPTION

Even though the p-value is statistically significant, concluding around 9.2% of variance in trophies depends on Best Brawler Type can still be misleading due to uneven sample sizes of players across brawler types.

2. Kruskal-Wallis Test

```
Best Brawler:
Kruskal-Wallis H: 64.98505697178433
p-value: 0.37313588674800496
Epsilon squared: 0.021788737020323586
```

STATISTICALLY INSIGNIFICANT

The variable Best Brawler remains statistically insignificant even under the Kruskal-Wallis test, which does not assume normality.

```
Best Brawler Type:
Kruskal-Wallis H: 17.79776244129679
p-value: 0.022794863073539715
Epsilon squared: 0.051297185556527694
```

STATISTICALLY SIGNIFICANT

This result is more reliable than the ANOVA output.

However, the practical impact is modest, indicating Best Brawler Type has an effect of **5.1%** on trophy count, which is to be interpreted with caution.

BORDA COUNT

Borda points by Best Brawler Type:

BestBrawlerType	
Assassin	4024
Damage Dealer	3422
Tank	2738
Controller	2430
Support	2429
Marksman	1976
Unknown	1324
Sharpshooter	1251
Artillery	306

KEY FINDINGS

- The Borda Count rewards both frequency and placement across the table
- Despite Damage Dealers appearing more frequently than Assassins, **Assassins placed higher ranks** and collected more points

CONCLUSION AND APPLICATION

3V3 AND SOLO WINS

While players should primarily focus on 3v3 matches, maintaining strong individual performance through solo modes remains important for climbing the ranks. Designing more solo play modes may help foster more competitive players.

BRAWLER TYPES

Top players should become skilled utilizing Assassin-type brawlers in order to climb the ranks. The Artillery class showed a substantial performance deficit compared to the others, highlighting an opportunity for balance improvements.

THANK YOU

Project Repository: github.com/zhamanta-chavez/Brawl-Stars-Analysis

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